

Module 2

1. Consider the following schemas [6]
Emp(Emp_no, Emp_name, Dept_no)
Dept(Dept_no, Dept_name)
Address(Dept_name, Dept_location)
Write SQL queries for the following
 - i) Display the location of department where employee 'Ram' is working.
 - ii) Create a view to store total no of employees working in each department in ascending order.
 - iii) Find the name of the department in which no employee is working.
2. Consider following schema
Hotels(hotel_no,hotel_name,city)
Rooms(Room_no,hotel_no,price,type)
Write a PL/SQL procedure to list the price & type of all rooms at the hotel 'TAJ'.
3. Consider the following relation schema

MOVIES(Mov_Id, Mov_Title, Mov_Year, Dir_Id)
DIRECTOR(Dir_Id, Dir_Name)
RATING(MOV_Id, Rev_Stars)
Write the SQL queries for the following.
 - i. List the title of all the movies directed by 'RAJ KAPOOR'
 - ii. Find the name of movies and number of stars for each movie. Sort the results on movies title and from higher stars to least stars.
 - iii. Assign the rating of all movies directed by 'Steven Spielberg' to 9.
4. What is the importance of creating constraints on the table? Explain with example any 4 constraints that can be specified when a database table is created.
5. Explain the concept of Referential and Entity Integrity constraint with example.
6. Defined stored procedure. Explain the creating and calling stored
7. procedure with example.
8. c) Consider following schema.
Student_fee_details (rollno, name, fee_deposited, date)
Write a trigger to preserve old values of student fee details before updating in the table.
9. What is an index? What are the advantages and disadvantages of using index on a table?

10. Explain cursor in SQL with example.
11. What is a trigger? How to create it? Discuss various types of triggers.
12. Write a PL/SQL block of code which accepts the rollno from user. The attendance of rollno entered by user will be checked in student_attendance(RollNo, Attendance) table and display on the screen.
13. What is view and how to create it? Can you update view? If yes, how? If not, why not?
14. What is join operation in SQL? Explain any two types of joins with example.
15. Consider following schema
 Student_fee_details (rollno, name, fee_deposited, date)
 Write a trigger to preserve old values of student fee details before updating in the table.
16. Defined stored function. Explain the creating and calling stored function with example.
 Explain set operations, set membership operations with example.
17. Consider the following schemes [6]
 Supplier(SNO, Sname, Status, City)
 Parts (PNO, Pname, Color, Weight, City)
 Shipments(SNO,PNO,QTY)
 Write SQL queries for the following:
 i) Find shipment information (SNO, Sname, PNO, Pname, QTY) for those having quantity less than 157.
 ii) List SNO, Sname, PNO, Pname for those suppliers who made shipments of parts whose quantity is larger than the average quantity
 iii) Find aggregate quantity of PNO 1692 of color green for which shipments made by supplier number who residing Mumbai

Module 3

1. Explain document based data model of NOSQL database.
2. Explain Graph based data model of NOSQL database.
3. Explain column based data model of NOSQL database.
4. Explain key-value based data model of NOSQL database.
5. Explain BASE properties of NOSQL database with suitable example.
6. Explain aggregation in MongoDB with suitable example.
7. Explain CAP theorem in NOSQL database.
8. Explain following NOSQL database types with examples and also state
9. the scenario where it is useful [9]
 - i) Column-oriented
 - ii) Graph

- iii) Document -oriented
10. Explain CAP theorem and BASE properties.
 11. Differentiate between relational database and NOSQL database.
 12. List different NOSQL data models and explain any one in detail with example.
 13. List the different NOSQL data models. Explain following NOSQL database types with examples:
 - i. Column-oriented
 - ii. Document -oriented
 14. Explain the NOSQL database types with examples and write down the real time applications.
 15. Explain the CAP theorem referred during the development of any distributed application.
 16. Consider the following Movie collection

```
{ "title": "Dangal", "genre": "Drama", "language": "Hindi", "releaseYear": 2016,
"rating": 8.4, "boxOffice": 2024000000, "cast": [ { "actor": "Aamir Khan", "role":
"Mahavir Singh Phogat" }, { "actor": "Fatima Sana Shaikh", "role": "Geeta Phogat" }
]}
```

Write MongoDB queries for the following:

- i) Count Hindi Movies in Each Genre
 - ii) Find Average Rating of Hindi Movies by Genre
 - iii) List Hindi Movies Released After 2015
17. Explain how aggregation works in MongoDB. Use aggregate function to find the customer who has given highest business for the product “toothbrush”, consider following collections:

```
db.purchase_orders.insertMany( [
{product: "toothbrush", total: 4.75, customer: "Mike"},
{product: "guitar", total: 199.99, customer: "Tom"},
{product: "milk", total: 11.33, customer: "Mike"},
{product: "pizza", total: 8.50, customer: "Karen"},
{product: "toothbrush", total: 4.75, customer: "Karen"},
{product: "pizza", total: 4.75, customer: "Dave"}
{product: "toothbrush", total: 4.75, customer: "Mike"}] ) ;
```

Module 4

1. Define transaction management. Explain ACID properties with suitable example.
2. Explain different transaction state.
3. State and explain the ACID Properties with example. During its execution, a transaction passes through several states, until it finally commits or aborts. List all possible sequences

of states through which a transaction may pass. Explain the situations when each state transition occurs

4. Explain time stamp based and log-based concurrency control protocol
5. What is serializability? Explain conflict serializability with a suitable example.
6. Explain lock-based concurrency protocol in database.
7. Check whether given schedule is conflict serializable, view serializable. Justify your answer.

T3	T4	T5
Read(Q)		
	Write(Q)	
Write(Q)		
		Write(Q)

8. What is conflict serializability? How to check schedule is conflict serializable schedule? Explain with one example.
9. Explain two phase locking protocol. Consider the following two transactions:

T1: read(A);
 read(B);
 if A = 0 then B:=B+1;
 Write (B).

T2: read(B);
 read(A);
 if B= 0 then A: = A+1; write (A).

Add lock and unlock instructions to transactions T1 and T2, so that they observe the two phase locking protocol. Can the execution of these transactions result in a deadlock?

10. How to ensure the atomicity using Recovery Methods? Explain the log based recovery method in detail.
11. What is R-timestamp(Q) and W-timestamp(Q). Explain the necessary condition used by time stamp ordering protocol to execute for a read /write operation.
12. What is need of lock in DBMS? Explain shared lock and exclusive lock with the help of example.
13. A transaction may be waiting for more time for an Exclusive (X) lock on an item, while a sequence of other transactions request and are granted as Shared (S) lock on the same item. What is this problem? How is it solved by two phase lock protocol?
14. Explain with example, recoverable and non recoverable, cascadeless schedules.
15. Check if given schedule is recoverable or non recoverable. Justify your answer.

T1	T2
	read (A)
read (B)	
Read (A)	
write (A)	
	write (B)
	commit
Abort	

16. Assume that initially: $R-TS(A) = W-TS(A) = 0$ $R-TS(B) = W-TS(B) = 0$

Assume $TS(T1) = 1$, $TS(T2) = 2$

What will be the output of the basic timestamp-based protocol for the given schedule?

T1	T2
	read (A)
read (B)	
Read (A)	
write (A)	
	write (B)

17. What is conflict serializability? Check following schedule is conflict serializable or not? Also, explain the concept of conflict equivalent schedule.

T1	T2	T3	T4
R(X)			
R(Z)			
	W(X)		
		R(Y)	
		W(Y)	
			W(X)
			W(Y)
			W(Z)

$R(X)$ denotes read operation on data item X by transaction T_i .

$W(X)$ denotes write operation on data item X by transaction T_i .